

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

### COURSE OUTCOME COVERED

*CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)*

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

### Annexure A

#### Analytical Problem Solving

- A coin is tossed 10 times and the outcomes are:  
H H T H H H T H H T  
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
- Out of 50 students, 42 passed an exam.  
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
- A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.
- A biased die is rolled 60 times. The number 6 appears 18 times.  
Estimate the probability of rolling a 6 using MLE.
- Given:  
Input (x) = 4  
Weight (w) = 2  
Actual Output = 10  
Prediction:  $y = w \times x$
- Find the predicted output.
- Calculate the error.

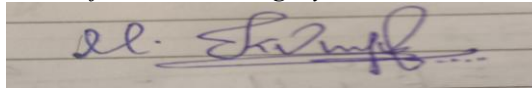
- If the gradient is 4 and learning rate is 0.01, find the updated weight.

*Code of Conduct:*

*I Kumaran.M(192525022) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*



#### RUBRICS FOR CALCULATION

| Criteria              | Weightage | Level 4 – Exemplary (5 Marks)                                                     | Level 3 – Proficient (4 Marks)           | Level 2 – Developing (2–3 Marks)           | Level 1 – Beginning (0–1 Mark)                |
|-----------------------|-----------|-----------------------------------------------------------------------------------|------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Problem Understanding | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps      | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection      | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors | Inappropriate or partially correct method  | Unable to select method                       |
| Solution Process      | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors        | No clear process                              |

|                           |     |                                                                 |                                           |                                         |                       |
|---------------------------|-----|-----------------------------------------------------------------|-------------------------------------------|-----------------------------------------|-----------------------|
| Accuracy of Calculation   | 20% | All calculations accurate; correct final answer                 | Minor calculation errors                  | Frequent errors; incorrect final answer | No valid calculations |
| Interpretation of Results | 15% | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation       | No interpretation     |

**solutions:**

1. Coin Toss – Maximum Likelihood Estimation (MLE) Given outcomes:

H H T H H H T H H T

Total tosses,  $n=10$

Number of Heads = 7

The MLE for the probability of Heads is:

$p = \text{Number of Heads} / \text{Total Tosses}$

$p = 7/10 = 0.7$

**Answer:**  $p=0.7$

2. Students Passing an Exam

**Given:**

Total students = 50

Passed = 42 MLE:

$p = 42 / 50 = 0.84$

**Answer:**

Probability of passing = 0.84 (84%)

### 3. Defective Light Bulbs

**Given:**

Total bulbs = 100

Defective bulbs = 8

MLE:

$$p = 8 / 100 = 0.08$$

**Answer:**

Probability of a bulb being defective = 0.08 (8%)

### 4. Biased Die **Given:**

Total rolls = 60

Number of times 6 appears = 18 MLE:

$$p = 18 / 60 = 0.3$$

**Answer:**

Probability of rolling a 6 = 0.3 (30%)

### 5. Prediction, Error and Weight

Update **Given:**

Input,  $x=4$

Weight,  $w=2$  Actual

Output = 10

Prediction formula:

$$y = w * x$$

(i) Predicted Output

$$y = 2 \times 4 = 8$$

**Answer:**

$$\text{Predicted Output} = 8$$

(ii) Calculate the Error

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

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$$= 10 - 8 = 2$$

**Answer:**

$$\text{Error} = 2$$

(iii) Updated Weight **Given:**

$$\text{Gradient} = 4 \times \text{Learning Rate}$$

$$= 0.01 \times \text{Gradient descent}$$

update rule:

$$w(\text{new}) = w - (\text{Learning Rate} \times \text{Gradient})$$

$$= 2 - (0.01 \times 4)$$

$$= 2 - 0.04$$

$$= 1.96$$

**Answer:**

$$\text{Updated Weight} = 1.96$$



